Flow in Wind Tunnels

AA 504 (Compressible Fluid Mechanics) Autumn 2026, UW

Assignment: Homework #2

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Problem 1. Stagnation Pressure on a Model in a Supersonic Tunnel

Problem Statement

Determine the stagnation pressure on a wind tunnel model in a “blow down” supersonic tunnel. The test section Mach number is 4 and the stagnation pressure upstream of the choked throat is 20 atm. The stagnation temperature is 300K.

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Analysis

The normal-shock relations for a perfect gas are used:

$$M_2^2 = \frac{1 + \frac{\gamma - 1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma - 1}{2}}, \quad \text{(2.45) [3]}$$

and

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1) \quad \text{(2.48a) [3]}$$

from the analysis performed in Problem 2, we have $p_1 = 0.1317 atm$ the static pressure ahead of the shock

→ **p_2 = 18.5(0.1317 atm)** $= 2.4365\ atm$

Using (2.45) to solve for the Mach speed immediately following the shock:

→ **M_2 = 0.4350**

Now we bring the post-shock air to rest on the model:

- Since we're assuming isentropic flow - the flow is adiabatic and in (local thermodynamic) equilibrium, we use (2.31) to find the stagnation pressure on the model.

$$\frac{p_{02}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)}$$
$$= 1.1388$$

→ **p_02 = 1.1388(2.4365 atm) = 2.775 atm**

- Only ~13.9% (2.775/20) of the reservoir stagnation pressure remains after the shock.

Note: Stagnation temperature tracks total energy, while stagnation pressure depends on both energy and entropy. -- A shock conserves total energy but generates entropy, which is why the two behave differently.

Problem 2. Static Pressure and Temperature

Problem Statement

What are the static pressure and temperature in an empty test section of the tunnel in problem 1?

Analysis

Begin with first principles; **conservation of energy.**

Heat added + Work done = change in Energy

$$q + w = \Delta E$$

- $q = 0$; Isentropic flow -- constant entropy, adiabatic (no heat transfer) and reversible
- $w = 0$; there is no work being done on the test section (CV) by any kind of device (turbine, pump, etc.)

$0 = \Delta h_{total}$; total enthalpy

- $\Delta h_{total} = h_0 + \frac{u_0^2}{2} - h_1 - \frac{u_1^2}{2}$
- total enthalpy is the change in enthalpy measured at two equilibrium end points (station 0 [the reservoir] and station 1)

Specific Heat Relation

For a thermally perfect gas (satisfies the ideal gas law: $p = \rho RT$) and if it is also calorically perfect, $c_p = const.$, then $h = c_p T$. [3]

Stagnation Properties

The velocity of the flow is essentially zero at the reservoir, we have:

$$h_0 = h_{stagnation} = c_p T_0 = c_p T_1 + \frac{u_1^2}{2}$$

- $T_1 = T_{static}$; the static temperature.

Perfect Gas Isentropic Relations

Speed of sound: $a^2 = \gamma RT$

Mach number: $M = \frac{u}{a} \quad \rightarrow u = Ma = M\sqrt{\gamma RT}$

Specific gas constant: $R = c_p - c_v$

Heat capacity ratio: $\gamma = \frac{c_p}{c_v}$

- $c_p = \frac{\gamma R}{\gamma - 1}$

These isentropic relation (and a whole bunch of algebra!) gives:

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad (2.30)$$ [3]

At $T_0 = 300K$:

T = T_static = 71.43 K

The nozzle converts enthalpy into kinetic energy, which lowers the static temperature.

Similarly, from the isentropic relations, we know:

$$\frac{p_{01}}{p_1} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)} \quad \text{(2.31) [3]}$$

p_1 is the static pressure in the empty test section of the tunnel.

→ p_1 = 0.1317 atm

Problem 3. Density Ratio Across a Shock Wave

Problem Statement

Determine the density ratio across a shock wave in the limit of infinite strength.

Analysis

Using the density ratio relation (2.47) [3] will give the compression of the gas to its upstream value.

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2} \quad \text{(2.47)}$$

An infinitely strong normal shock corresponds to $M_1 \to \infty$;

- Apply some analytical tactics to delete two M_1 variables from the density ratio relation:

Take the infinite strength limit:

-divide the numerator and denominator by M_1^2:

$$= \frac{(\gamma + 1)}{\frac{2}{M_1^2} + (\gamma - 1)}$$

As $M_1 \to \infty$, $\frac{2}{M_1^2} \to 0$, giving:

$$\lim_{M_1 \to \infty} \frac{\rho_2}{\rho_1} = \frac{\gamma + 1}{\gamma - 1}$$

For air: $\gamma = 1.4$

$$\frac{\rho_2}{\rho_1} \longrightarrow \frac{2.4}{0.4} = \mathbf{6}$$

- **A shock wave with infinite strength compresses the gas to six-times its upstream value. Note: This only applies to the constant-γ ideal gas model.**
- Real air at sufficiently high-temperatures changes its molecular behavior, so six is not a universal physical limit.

Problem 4. Rarefied Gas Dynamics

Problem Statement

Based off of the movie "Rarefied Gas Dynamics" [1], write one paragraph on the most surprising aspect of the movie to you.

Statement

The central message the movie presents is that gas behaves differently depending on how far its molecules travel between collisions compared with the size of the object or flow being studied. This determines whether familiar continuum concepts apply to these types of gases.

Some key concepts include:

1. Mean free path measures the distance between molecular collisions.

The mean free path, \(\lambda\), is the average distance a molecule travels between collisions with other gas molecules.

As gas density decreases, collisions become less frequent and the mean free path increases. Consequently, a gas at high altitude or inside a vacuum chamber can behave very differently from air at sea level.

2. The Knudsen number determines the importance of rarefaction.

The relevant quantity is the mean free path relative to a characteristic length, L:

$$\mathrm{Kn} = \frac{\lambda}{L}$$

Regime	Relative length scales	Behavior demonstrated
Continuum flow	$\lambda \ll L$	Frequent molecular collisions produce familiar flow structures, including well-defined shocks.
Transitional flow	λ is comparable to L	Collisions matter, but continuum descriptions become less reliable. Shocks become diffuse.
Free-molecular flow	$\lambda \gg L$	Molecules largely follow independent trajectories; collisions with solid surfaces dominate their interaction with a body.

The choice of L matters. In the presentation, a tiny wire experiences free-molecular flow while a much larger model in the same stream experiences approximately continuum flow.

3. Shock waves have a finite physical thickness.

As collisions become more frequent, incoming molecules interact strongly with molecules scattered back from the model. A compressed region and a recognizable bow shock develop.

The experiments show that shock thickness is approximately proportional to the upstream mean free path: $\delta \propto \lambda$

At lower pressure, the mean free path increases and the shock becomes thicker. When its thickness is very small compared with the body, treating the shock as a discontinuity is a useful approximation.

This connects directly to our work on shocks ahead of wind tunnel models: **the thin shock used in conservation-law calculations represents a physical region where molecular collisions redistribute momentum and energy.**

Some applications of these principles:

Application	Why the presentation matters
High-altitude aerodynamics	Surface scattering and accommodation influence aerodynamic forces and heat transfer when intermolecular collisions are infrequent.
Wind tunnel testing	Matching Mach number alone does not establish comparable rarefaction conditions; the relevant Knudsen number also matters.

References

[1] YouTube '17. Rarefied Gas Dynamics'.

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] Liepmann, H. W., & Roshko, A. (2001). *Elements of gasdynamics*. Dover Publications. (Original work published 1957)